

Assignment 1 - GAN Playground

Deep Neural Networks for Computer Graphics

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Task 1.2a - Padding Derivation

We use a 2D convolution with kernel size $K = 4$ and stride $S = 2$ to downsample the spatial dimension by a factor of 2, meaning we require $H_{out} = \frac{H_{in}}{2}$. The standard output-size formula for a convolutional layer is:

$$H_{out} = \lfloor \frac{H_{in} + 2P - K}{S} \rfloor + 1$$

Assuming perfect division (dropping the floor operator for even spatial dimensions typical in GANs), we substitute $K = 4$, $S = 2$, and $H_{out} = \frac{H_{in}}{2}$:

$$\frac{H_{in}}{2} = \frac{H_{in} + 2P - 4}{2} + 1$$

Multiplying the entire equation by 2:

$$H_{in} = H_{in} + 2P - 4 + 2$$

$$H_{in} = H_{in} + 2P - 2$$

$$0 = 2P - 2$$

$$2P = 2$$

$$P = 1$$

Thus, the required padding is $\mathbf{P = 1}$.

Weights & Biases Reports

Below are the links to our public W&B reports containing the required loss curves, generated samples, and qualitative discussions for each task.

- **Task 1.7 - DCGAN W&B Report:**
[Click here to view the DCGAN Report](#)
- **Task 2.6 - CycleGAN W&B Report:**
[Click here to view the CycleGAN Report](#)
- **Task 2.7 Bonus - Image Replay Buffer W&B Report:**
[Click here to view the Bonus Report](#)